TED (21) – 2022 REVISION 2021

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING AND TECHNOLOGY

MANUFACTURING TECHNOLOGY

MODEL QUESTION PAPER – SET-1

Time: 3 hours Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

1	Name any one mechanical property of an engineering material	M 1.01	U
2	__________is the percentage composition of clay in green sand?	M 1.03	U
3	Write any one hot working operation	M 2.04	U
4	Write an example for cold working operation	M 2.03	U
5	Write any one metal joining operation	M 3.01	U
6	Which welding process uses non consumable electrodes.	M 3.02	U
7	Write any one forging tool	M4.01	U
8	Anvil is specified by its ________	M 4.02	U
9	Write any one press working operation.	M 4.02	U

PART B

II. Answer any Eight questions from the following

(8 x 3 = 24 Marks)

1	Explain any three mechanical properties of engineering materials	M 1.01	U
2	Explain types of patterns used in casting process	M 1.02	U
3	Explain about the ingredients in moulding sand	M 1.03	U
4	Explain centrifugal casting	M 2.01	U
5	Explain any one hot working operation	M 2.04	U
6	Explain the classification of fusion welding process	M 3.01	U
7	Explain arc welding process	M 3.03	U

8	Explain briefly about welding electrodes	M 3.04	U
9	Explain flat die forging	M 4.01	U
10	List various press working operations	M 4.03	U

PART C

III. Answer all questions from the following

(6 x 7 = 42 Marks)

1	Explain various pattern allowances with neat sketches	M 1.02	U
	OR		
2	Explain different types of moulding process	M 1.04	U
3	Explain different types of die casting with sketches	M 2.01	U
	OR		
4	Explain the advantages and disadvantages of cold and hot working operations	M 2.03	U
5	Explain various defects in casting with simple sketches	M 2.02	U
	OR		
6	Explain briefly the various steps in powder metallurgy process	M 2.05	U
7	Explain different types of welding flames	M 3.02	U
	OR		
8	Explain TIG welding process	M 3.03	U
9	Explain about various welding positions with sketches	M 3.04	U
	OR		
10	Compare brazing and soldering	M 3.05	U
11	Explain different types of forging operations	M 4.01	U
	OR		
12	Compare punching and blanking operations	M 4.04	U

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MANUFACTURING TECHNOLOGY

MODEL QUESTION PAPER – SET-2

Time: 3 hours Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

1	Property of a material which can be drawn into thin wires under the action of tensile forces is called	M 1.01	U
2	Property of moulding sand to withstand high temperatures is_______	M 1.03	U
3	Write any one casting defect	M 2.02	U
4	Name any one cold working operation	M 2.03	U
5	Write an example for pressure welding	M 3.01	U
6	Heat is created by chemical reaction in ________ welding	M 3.02	U
7	Which processes is mainly used for making the connecting rods?	M4.01	U
8	Write any one forging tool used in forging process	M 4.02	U
9	In blanking operation, the clearance is provided on the die for obtaining the blank from metal sheet. (T/F)	M 4.04	U

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

1	Explain the significance of manufacturing process in engineering industry	M 1.01	U
2	Discuss the advantage and disadvantage of wood pattern material	M 1.02	U
3	Explain any three properties of moulding sand	M 1.03	U

4	Explain slush casting with an example	M 2.01	U
5	Explain any three defects in castings	M 2.02	U
6	Explain briefly about different types of gas welding techniques	M 3.02	U
7	Explain thermit welding technique	M 3.03	U
8	Write any one forging tool used in forging process	M 4.02	U
9	Explain closed die forging	M 4.01	U
10	Explain the significance of press working operations	M 4.03	U

PART C

III. Answer all questions from the following

(6 x 7 = 42 Marks)

1	Explain different types of pattern with simple sketches	M 1.01	U
	OR		
2	Explain the gating system used in casting process with sketches	M 1.02	U
3	Explain goose neck type casting method with neat sketches	M 2.01	U
	OR		
4	Briefly explain the different hot working operations	M 2.04	U
5	Explain bending and shearing operation with sketches	M 2.03	U
	OR		
6	Explain the advantages and applications of powder metallurgy	M 2.03	U
7	Explain AC and DC welding machines	M 3.01	U
	OR		
8	Explain shielded metal arc welding technique	M 3.03	U
9	Explain MIG welding process	M 3.03	U
	OR		
10	Explain welding electrodes and functions of electrode coatings	M 3.04	U

11	Explain about various forging tools	M 4.01	U
OR			
12	Explain various press working operations	M 4.03	U